ApproxDA-TransUNet: A highly efficient medical image segmentation method with dual spatial and channel attention approximation

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Abstract

Dual-attention networks capture spatial and channel-wise dependencies for medical image segmentation, but models such as DA-TransUNet are expensive (≈108M parameters, ≈30 GFLOPs per 224×224 slice) because of full-map self-attention, a ViT-B/16 encoder, and a wide decoder. We present **ApproxDA-TransUNet**, a lightweight architecture that combines a ResNet-50 stem truncated at stride 8, a LightViT-T bottleneck transformer, dual-attention approximation blocks (local windowing $M = 28$, low-rank projection $r = 32$, grouped channels $G = 8$) on the bottleneck and all three skip connections, and a slim decoder with 64-channel skip stages. On five benchmarks (Synapse CT, ACDC MRI, CVC-ClinicDB colonoscopy, ISIC 2018 dermoscopy, and FeTA 2.4 fetal MRI), each over three seeds, ApproxDA-TransUNet uses 90.0% fewer parameters (10.75M vs. 107.95M) and 77.1% fewer FLOPs (6.92 vs. 30.21 GFLOPs) than DA-TransUNet while matching or slightly exceeding its Dice: the mean paired gain over 15 seed pairs is +0.37 percentage points (10/15 positive, Wilcoxon $p = 0.010$), although no single dataset reaches significance. On FeTA 2.4 it matches TransUNet within 0.1 points at one tenth of the parameters. Ablations show that the ApproxDA block is necessary (-2.76 points without it, $p = 0.008$), whereas the choice of attention operator inside it (approximate, exact, or none) does not change accuracy. A gradient analysis explains this: with zero-initialized residual

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scales, SGD leaves the position-attention weights essentially untrained.

Keywords: Biomedical image segmentation, Lightweight architecture, Dual-attention approximation, Vision Transformer, Multi-organ segmentation

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1. Introduction

Automated biomedical image segmentation is fundamental to computer-assisted diagnosis, radiotherapy planning, and surgical navigation [1, 2, 3]. Clinical modalities span a broad spectrum—from 3D volumetric CT and MRI to 2D endoscopic video and high-resolution dermoscopy. Accurate boundary delineation across these domains requires architectures capable of both fine-grained edge localization and expansive global context modeling.

While convolutional neural networks (CNNs) [1, 2, 4] form the conventional baseline, their localized receptive fields limit long-range context aggregation. Hybrid CNN-Transformer frameworks like TransUNet [3] address this via self-attention, and dual-attention networks [5] concurrently capture spatial affinities via Position Attention Modules (PAM) and cross-channel correlations via Channel Attention Modules (CAM). Recently, DA-TransUNet [6] integrated dual-attention blocks across multi-scale skip connections and the bottleneck, demonstrating state-of-the-art accuracy on multi-organ CT and colon polyp benchmarks.

Despite their representational strength, standard dual-attention architectures suffer from severe computational overhead. Full-map PAM scales quadratically with spatial tokens ($\mathcal{O}(N^2C)$), while CAM scales quadratically with channel depth ($\mathcal{O}(C^2N)$). Coupled with a ViT-B/16 encoder and a wide decoder, DA-TransUNet requires **107.95M parameters** and **30.2 GFLOPs** per 224×224 slice (39.8 GFLOPs at 256×256), which limits its use on resource-constrained hardware.

In this work, expanding upon our preliminary workshop exploration [7], we present **ApproxDA-TransUNet**, a redesigned lightweight architecture built from four components: (i) a *ResNet-50 stem truncated after its second stage* (stride 8, 512 channels) followed by a compact Vision Transformer (*LightViT-T* [8]), which reduces the encoder from 97.9M to 9.3M parameters; (ii) a *tri-axial attention approximation* parameterizing spatial windowing ($M = 28$), shared low-rank subspace projection ($r = 32$), and grouped channel attention ($G = 8$) with fixed equal-blend gating ($g = 0.5$); (iii) *multi-*

scale skip modulation embedding ApproxDA blocks uniformly across all skip connections $(1/8, 1/4, 1/2)$ and the bottleneck; and (iv) a *slim decoder* with additive skip fusion and 64 channels at every skip stage. Together, these changes reduce the parameters from **107.95M to 10.75M** (-90.0%) and the compute from **30.21 to 6.92 GFLOPs** (-77.1%).

We evaluate ApproxDA-TransUNet across five multi-modal benchmarks under a strict three-seed protocol (1234, 0, 1): Synapse CT, ACDC cardiac MRI, CVC-ClinicDB colonoscopy, ISIC 2018 dermoscopy, and FeTA 2.4 fetal brain MRI. Across the 15 seed pairs, ApproxDA-TransUNet matches or slightly exceeds DA-TransUNet (mean paired Dice gain +0.37 percentage points, $10/15$ pairs positive, Wilcoxon $p = 0.010$); no single dataset is significant on its own, and HD95 is higher than DA-TransUNet's on four of the five datasets. On FeTA 2.4, the 10.75M model is 1.8–2.5 points above the reproduced U-Net family and Swin-UNet and within 0.1 points of TransUNet. Ablations on Synapse show that the ApproxDA block is necessary but that the attention operator inside it can be approximated, made exact, or removed without changing accuracy; a gradient analysis of the zero-initialized residual scales explains why.

The primary contributions of this paper are:

- **Lightweight dual-attention architecture**: ApproxDA-TransUNet uses 90% fewer parameters and 77% fewer FLOPs than DA-TransUNet at comparable segmentation accuracy.
- **Attention approximation**: a window (M), low-rank (r), and channel-group (G) approximation of the dual-attention block, with ablations showing that it costs no accuracy relative to exact dual attention.
- **Multi-modal three-seed evaluation**: paired comparisons on five benchmarks spanning CT, MRI, colonoscopy, and dermoscopy.
- **Optimization diagnostics**: an analysis of zero-initialized residual scales in dual attention under SGD, which explains why the block's convolutional path, rather than its attention matrices, carries its contribution.

The rest of this paper is organized as follows: Section 2 reviews related work; Section 3 presents the methodology; Section 4 reports experimental results; Section 5 provides ablation studies; Section 6 discusses optimization diagnostics and deployment profiles; Section 7 concludes the paper.

2. Related Work

2.1. Architectures for Biomedical Image Segmentation

U-Net [1] established the dominant encoder-decoder topology for medical image segmentation, using multi-scale skip connections to preserve high-resolution spatial details. Subsequent variants enhanced representation capacity through nested dense skips (UNet++ [2]), additive attention gates (Attention U-Net [4]), and multi-resolution residual blocks (MultiResUNet [9]). However, purely convolutional models remain constrained by localized receptive fields, hindering the disambiguation of contiguous anatomical structures.

To overcome this, Vision Transformers (ViTs) [10] were adopted for global context aggregation. TransUNet [3] pioneered hybrid architectures by coupling a ResNet-50 CNN with a ViT-B/16 bottleneck transformer. Swin-UNet [11] explored a purely transformer-based U-shaped structure with shifted local windows, while UNETR [12] and SwinUNETR [13] extended transformers to volumetric 3D segmentation. While effective, these models often exceed 100M parameters, imposing prohibitive memory and computational overhead.

2.2. Dual-Attention Mechanisms and Computational Bottlenecks

Dual attention was formalized by Fu et al. [5] (DANet) to simultaneously model spatial dependencies via Position Attention Modules (PAM) and cross-channel interdependencies via Channel Attention Modules (CAM). In biomedical imaging, DA-TransUNet [6] embedded dual-attention blocks across the bottleneck and skip connections of TransUNet, establishing state-of-the-art results on multi-organ CT and polyp segmentation. Related efforts explored dual attention in hierarchical transformers (DAE-Former [14], TransDAE [15]) and state-space architectures (DAMamba [16]).

However, standard dual attention exhibits quadratic complexity: spatial self-attention scales as $\mathcal{O}(N^2C)$ for $N = H \times W$ tokens, while channel attention scales as $\mathcal{O}(C^2N)$. In multi-scale architectures, generating full affinity matrices across multiple high-resolution skip stages (112×112 and 56×56) makes memory grow quadratically with the number of tokens.

2.3. Efficient Attention Approximation and Positioning

To mitigate quadratic complexity, diverse approximation paradigms have emerged in general vision: local window partitioning (Swin Transformer [17]),

low-rank key-value projection (Linformer [18], Performer [19]), channel group factorizations (EMCAD [20]), and efficient transformer backbones (LightViT [8]).

While prior works apply isolated approximation techniques, they retain heavy backbones (such as 86M ViT-B encoders) and high-channel decoders. In our preliminary workshop study [7], we analyzed spatial window sensitivity within the heavy ViT-B baseline. In contrast, this journal work delivers a holistic, system-level lightweight architecture: an early-truncated CNN stem, LightViT-T token processing, multi-scale tri-axial ApproxDA blocks, and a slim $C = 64$ decoder, and shows that a 90% parameter reduction is possible at comparable accuracy.

3. Methodology

This section describes ApproxDA-TransUNet: the overall topology, the attention approximation in the ApproxDA block, the slim decoder, and the computational cost.

3.1. Overall Lightweight Architecture

Figure 1 shows the complete architecture. Compared with the 108M-parameter DA-TransUNet [6], every stage is redesigned for efficiency while keeping the multi-scale skip structure.

3.1.1. Early-Truncated CNN Stem

In TransUNet and DA-TransUNet, the hybrid encoder runs the first three stages of ResNet-50 (down to stride 16 and 1024 channels; 11.9M parameters, 4.0 GFLOPs). The resulting 14×14 feature map is projected to 768-dimensional tokens and processed by a 12-layer ViT-B/16 (85.1M parameters, 17.4 GFLOPs).

ApproxDA-TransUNet truncates ResNet-50 after its second stage. For an input image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ (typically 224×224 or 256×256):

- **Stem & Stage 1**: Generates feature maps $\mathbf{F}_{1/2} \in \mathbb{R}^{64\times \frac{H}{2}\times \frac{W}{2}}$ (stride 2) and $\mathbf{F}_{1/4} \in \mathbb{R}^{256\times \frac{H}{4}\times \frac{W}{4}}$ (stride 4).
- **Stage 2**: Produces feature map $\mathbf{F}_{1/8} \in \mathbb{R}^{512\times \frac{H}{8}\times \frac{W}{8}}$ (stride 8).

The third stage is removed, so the CNN stem has 1.45M parameters (1.83 GFLOPs).

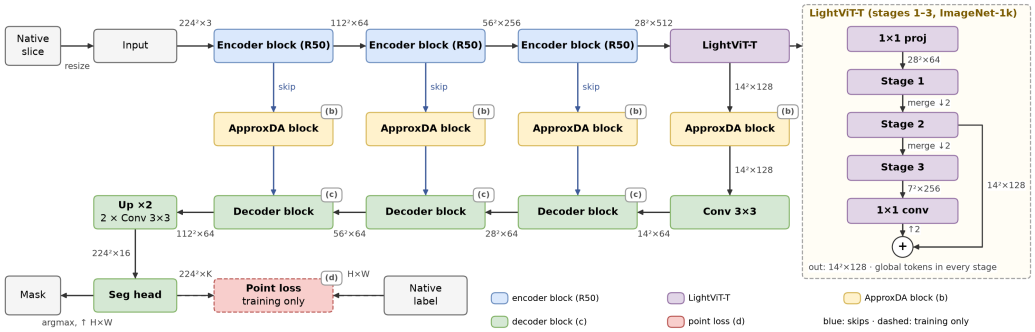


Figure 1: Overall architecture of the lightweight ApproxDA-TransUNet. The encoder consists of an early-truncated ResNet-50 (Stages 1–2, down to stride 8) followed by a top-down LightViT-T bottleneck. ApproxDA blocks are placed on all three skip connections (1/8, 1/4, 1/2) and on the bottleneck. The decoder uses 64 channels at every skip stage and 16 channels at full resolution (10.75M parameters in total). [TODO: the figure still shows the point loss (d) and the native label, which the final model no longer uses; regenerate it without them.]

3.1.2. LightViT-T Bottleneck and Attention Placement

Instead of ViT-B, $\mathbf{F}_{1/8}$ is projected by a 1×1 convolution and processed by the three stages of LightViT-T [8] (ImageNet-1k weights), a hierarchical lightweight vision transformer. The second-stage output and the upsampled third-stage output are fused into the bottleneck representation $\mathbf{Z} \in \mathbb{R}^{128 \times \frac{H}{16} \times \frac{W}{16}}$ (14×14 for 224×224 input). LightViT-T has 7.81M parameters but only 0.75 GFLOPs, because it operates on 28×28 and smaller token grids. The bottleneck ApproxDA block is positioned post-transformer, directly operating on $\mathbf{Z}$ to refine global representations before decoding.

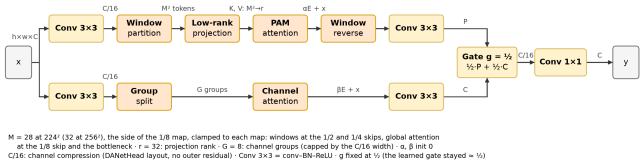
3.2. The ApproxDA Block Formulation

Figure 2 illustrates the internal topology of the ApproxDA block. The module receives input tensor $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ and splits computation into two parallel streams: Low-Rank Windowed Position Attention (PAM) and Grouped Channel Attention (CAM).

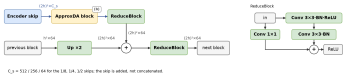
3.2.1. Bottleneck Convolutional Compression

As in the DANet head used by DA-TransUNet, each branch first compresses $\mathbf{X}$ with its own convolution:

$$\mathbf{X}' = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{X}))) \in \mathbb{R}^{C' \times H \times W}, \tag{1}$$



(a) ApproxDA Block with tri-axial approximation (M, r, G) and fixed gate $(g = 0.5)$.



(b) Slim Decoder Block $(C = 64)$.

Figure 2: Detailed architectural diagrams of (a) the ApproxDA block with Low-Rank Windowed PAM and Grouped CAM branches, and (b) the compact decoder block with uniform $C = 64$ channel width.

where $C' = C/16$. For the bottleneck and the 1/8, 1/4, and 1/2 skips, C' is 8, 32, 16, and 4.

3.2.2. Low-Rank Windowed Position Attention (PAM)

Full spatial self-attention requires pairwise affinities between all $N = HW$ tokens, yielding $\mathcal{O}(N^2C')$ complexity. We approximate this along two orthogonal dimensions: **spatial windowing** (M) and **low-rank projection** (r).

Window Partitioning. The feature map $\mathbf{X}'$ is partitioned into non-overlapping spatial windows of size $M \times M$, where $M = \min(M, H, W)$ ($M = 28$ for 224×224 inputs). Consequently, the 1/2 and 1/4 skip stages are partitioned into 4×4 and 2×2 window grids, while the 1/8 skip and bottleneck operate within a single global window. Within each window of $N_w = M^2$ tokens, queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$ are generated via 1×1 convolutions:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{X}_w, \quad \mathbf{K} = \mathbf{W}_K \mathbf{X}_w \in \mathbb{R}^{C_q \times N_w}, \quad \mathbf{V} = \mathbf{W}_V \mathbf{X}_w \in \mathbb{R}^{C' \times N_w}, \tag{2}$$

where $C_q = \max(1, \lfloor C'/8 \rfloor)$.

Low-Rank Subspace Projection. Rather than computing an $N_w \times N_w$ attention matrix, keys and values are projected into a subspace of rank r ($r = 32$) via shared projection matrix $\mathbf{P} \in \mathbb{R}^{N_w \times r}$:

$$\tilde{\mathbf{K}} = \mathbf{K}\mathbf{P} \in \mathbb{R}^{C_q \times r}, \quad \tilde{\mathbf{V}} = \mathbf{V}\mathbf{P} \in \mathbb{R}^{C' \times r}. \tag{3}$$

The windowed position attention matrix $\mathbf{A} \in \mathbb{R}^{N_w \times r}$ and refined features $\mathbf{E}_{\text{PAM}}$ are then computed as:

$$\mathbf{A} = \text{softmax}\left(\mathbf{Q}^\top \tilde{\mathbf{K}}\right), \tag{4}$$

$$\mathbf{E}_{\text{PAM}} = \alpha \left(\mathbf{A}\tilde{\mathbf{V}}^\top\right)^\top + \mathbf{X}_w \in \mathbb{R}^{C' \times N_w}, \tag{5}$$

where α is a learnable scalar initialized to 0. All window features are reassembled and passed through a 3×3 refinement convolution. Setting $r = 0$ bypasses projection, recovering exact attention within each window.

3.2.3. Grouped Channel Attention (CAM)

Standard CAM computes channel covariance across all C' channels with $\mathcal{O}(C'^2 N)$ complexity. ApproxDA blocks parameterize channel interactions via grouping parameter G: the C' channels are partitioned into G disjoint groups of width $C_g = C'/G$ (with $G \leftarrow \min(G, C')$):

$$\mathbf{X}' = \left[\mathbf{X}^{(1)}, \mathbf{X}^{(2)}, \ldots, \mathbf{X}^{(G)}\right], \quad \mathbf{X}^{(g)} \in \mathbb{R}^{C_g \times N}. \tag{6}$$

Within each group g, the channel correlation energy between channel slice i and j is $s_{ij} = \mathbf{X}_i^{(g)} \cdot \mathbf{X}_j^{(g)}$. The channel attention map and modulated features are given by:

$$x_{ij}^{(g)} = \text{softmax}_j\left(\max_k s_{ik} - s_{ij}\right), \tag{7}$$

$$\mathbf{E}_{\text{CAM},i}^{(g)} = \beta \sum_{j=1}^{C_g} x_{ij}^{(g)} \mathbf{X}_j^{(g)} + \mathbf{X}_i^{(g)}, \tag{8}$$

where β is a learnable scalar initialized to 0. Groups are concatenated along the channel dimension to form $\mathbf{E}_{\text{CAM}} \in \mathbb{R}^{C' \times H \times W}$ and refined by a 3×3 convolution. When G does not divide C', the largest divisor of C' not exceeding G is used. Because C' is small in the lightweight model, $G = 8$ leaves a single channel per group in the bottleneck ($C' = 8$) and in the 1/2 skip ($C' = 4$, four groups), where CAM reduces to $(1 + \beta)\mathbf{X}'$; the cost of CAM is negligible at every stage, and un-grouped CAM ($G = 1$) is evaluated in Section 5.

3.2.4. Fixed Equal-Blend Routing

In preliminary experiments, learnable gating vectors $\mathbf{g} = \sigma(\mathbf{W}_g\,\mathrm{GAP}(\mathbf{x}))$ systematically collapsed toward uniform blending ($g \approx 0.5$). Consequently, ApproxDA-TransUNet fixes gating to equal blending:

$$\mathbf{Y} = \mathbf{W}_f\,(0.5 \cdot \mathbf{E}_{\mathrm{PAM}} + 0.5 \cdot \mathbf{E}_{\mathrm{CAM}}) \in \mathbb{R}^{C\times H\times W}, \tag{9}$$

where $\mathbf{W}_f$ consists of channel dropout ($p = 0.05$), a 1×1 convolution back to C channels, and ReLU, as in the DANet head; there is no outer residual connection. [TODO: cite the evidence that the learned gate stayed near 0.5 (workshop gate analysis), or drop that claim.]

3.3. Compact Decoder Architecture

Unlike the concatenation decoder of DA-TransUNet ($256 \to 128 \to 64 \to 16$ channels), ApproxDA-TransUNet uses 64 channels at every skip stage and fuses skips by addition (Figure 2b). A 3×3 convolution first maps $\mathbf{Z}$ to 64 channels. At each skip stage $l \in \{1, 2, 3\}$ (1/8, 1/4, 1/2), the previous map is bilinearly upsampled by $2\times$ and added to the ApproxDA-refined skip $\mathbf{S}_l$ after a channel-reduction block:

$$\mathbf{D}_l = \mathrm{RB}\left(\mathrm{Upsample}_{2\times}(\mathbf{D}_{l-1}) + \mathrm{RB}(\mathbf{S}_l)\right) \in \mathbb{R}^{64\times H_l\times W_l}, \tag{10}$$

where RB is a residual block of two 3×3 convolutions with batch normalization and a 1×1 shortcut. The full-resolution stage upsamples once more and applies two 3×3 convolutions with 16 channels, followed by a 1×1 segmentation head. The decoder without the skip blocks has 0.95M parameters (3.68 GFLOPs).

3.4. Computational Complexity and Parameter Breakdown

As detailed in Table 1, ApproxDA-TransUNet reduces the parameters from 107.95M to **10.75M** (-90.0%) and the FLOPs from 30.21G to **6.92G** (-77.1%). Most of the saving comes from replacing ViT-B/16 and from the slim decoder; the attention blocks themselves are a small part of either model. For high-resolution skip features (112×112, $N = 12{,}544$), windowing ($M = 28$) and low-rank projection ($r = 32$) cut spatial affinity entries from $N^2 = 1.57 \times 10^8$ to $N \cdot r = 4.01 \times 10^5$ (a $392\times$ reduction). Channel grouping divides the size of the CAM affinity matrix by G, but CAM is already negligible after the $C/16$ compression.

Table 1: Parameters and FLOPs by component for DA-TransUNet and ApproxDA-TransUNet (224×224 input, 9 classes; fvcore on CPU, attention matrix products included). DA-TransUNet instantiates three DANet heads per decoder stage but executes one; its parameter count includes all of them.

Component	DA-TransUNet [6]		ApproxDA-TransUNet (Ours)	
	Params (M)	FLOPs (G)	Params (M)	FLOPs (G)
CNN stem	11.90 (R50, 3 stages)	3.96	1.45 (R50, 2 stages)	1.83
Token embedding	0.94	0.15	–	–
Bottleneck transformer	85.06 (ViT-B/16)	17.37	7.81 (LightViT-T)	0.75
Dual-attention blocks	2.68 (DANet heads)	0.87	0.54 (4 ApproxDA blocks)	0.60
Decoder (without attention blocks)	7.39	7.78	0.95	3.68
Segmentation head	< 0.01	0.07	< 0.01	0.07
Total	**107.95**	**30.21**	**10.75**	**6.92**
Relative	100%	100%	**10.0%**	**22.9%**

3.5. Optimization Objective

All models are trained with a composite objective combining Cross-Entropy ($\mathcal{L}_{\text{CE}}$) and multi-class soft Dice loss ($\mathcal{L}_{\text{Dice}}$) with equal weighting:

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{CE}} + 0.5 \cdot \mathcal{L}_{\text{Dice}}. \tag{11}$$

No auxiliary losses or extra heads are used.

4. Experiments and Results

This section details the experimental protocol across five diverse clinical benchmarks, reports multi-seed quantitative comparisons between ApproxDA-TransUNet and DA-TransUNet, presents the FeTA fetal brain benchmark against 2D baselines and 3D references, and shows qualitative results.

4.1. Experimental Setup and Benchmark Datasets

Models are evaluated across **five clinical benchmarks** spanning four distinct imaging modalities under a strict **three-seed protocol** (seeds: 1234, 0, 1):

1. **Synapse Multi-Organ CT** [21]: 30 volumetric abdominal CT scans (18 train, 12 test; 2211 slices) covering 8 abdominal organs (aorta, gallbladder, left/right kidneys, liver, pancreas, spleen, stomach), trained for 300 epochs at 224×224 and evaluated with Dice (%) and HD95 in voxels (the TransUNet evaluation code, without voxel spacing).

2. **ACDC Cardiac Cine-MRI** [22]: 100 patient exams (40 end-diastolic/systolic 3D test volumes) targeting right ventricle, myocardium, and left ventricle, trained for 300 epochs at 224×224 and evaluated with Dice (%) and HD95 (px).
3. **CVC-ClinicDB Colonoscopy** [23]: 612 colonoscopy frames (75/25 split) for binary polyp segmentation, trained with Adam (10^{-3}) for 500 epochs at 256×256, evaluated with Dice (%) and HD95 (px).
4. **ISIC 2018 Dermoscopy** [24]: 2594 dermoscopic images (75/25 split) for binary skin-lesion segmentation, trained with SGD for 50 epochs at 256×256, evaluated with Dice (%) and HD95 (px).
5. **FeTA 2.4 Fetal Brain MRI** [25]: 80 reconstructed T2-weighted fetal brain MRI volumes under the standard EffiDec3D split (64 train, 8 val, 8 test) across 7 brain tissues (eCSF, GM, WM, LV, CBM, SGM, BS), trained for 61 epochs at 224×224 and evaluated with Dice (%) and HD95 (mm, using each subject's voxel spacing).

Implementation Protocols. All models are trained with the TransUNet augmentations (random 90° rotations with flips, and small rotations), a $0.5\,\mathcal{L}_{\text{CE}} + 0.5\,\mathcal{L}_{\text{Dice}}$ loss, batch size 24, and SGD (learning rate 0.01, momentum 0.9, weight decay 10^{-4}, polynomial decay), except CVC-ClinicDB (Adam, 10^{-3}). DA-TransUNet and ApproxDA-TransUNet are trained under identical settings. Every model is evaluated at the *final training epoch*, without checkpoint selection. Volume metrics are computed per case and averaged over classes; on FeTA, classes absent from a ground-truth volume are excluded from the average, as in nnU-Net.

4.2. Main Multi-Seed Benchmark Results

Table 2 summarizes the results. With a tenth of the parameters and under a quarter of the FLOPs, ApproxDA-TransUNet has a higher mean Dice than DA-TransUNet on all five benchmarks: +0.32 points on Synapse (seed standard deviation 0.20 vs. 0.66), +0.15 on ACDC, +0.41 on CVC-ClinicDB, +0.20 on ISIC 2018, and +0.80 on FeTA 2.4 (positive for all three seeds: +0.73, +1.47, +0.19). None of these per-dataset differences is significant with three seeds.

Pooling the 15 seed pairs, 10 favor ApproxDA-TransUNet, with a mean gain of +0.37 **points** (Wilcoxon signed-rank $p = 0.010$; paired t-test $p = 0.007$). HD95 does not follow the Dice trend: it is lower than DA-TransUNet's

Table 2: Quantitative comparison across five clinical benchmarks evaluated over three independent random seeds (1234 / 0 / 1). DSC is reported as individual seed scores and Mean ± Standard Deviation. HD95 is reported in millimeters for FeTA, in voxels for Synapse, and in pixels for ACDC, CVC-ClinicDB, and ISIC 2018. p-values are paired t-tests over the three seeds.

Dataset / Model	Params (M)	FLOPs (G)	Dice (%) (Seed 1234 / 0 / 1)	Mean ± Std Dice (%)	HD95	Paired Diff. (vs. DA)
Synapse Multi-Organ CT (8 organs, HD95 in voxels)						
DA-TransUNet [6]	107.95	30.21	79.74 / 81.00 / 80.03	80.26 ± 0.66	**23.49**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	80.61 / 80.75 / 80.36	**80.57 ± 0.20**	24.52	+**0.32** (p=0.43)
ACDC Cardiac Cine-MRI (3 structures, HD95 in px)						
DA-TransUNet [6]	107.95	30.21	90.66 / 91.06 / 91.09	90.94 ± 0.24	**1.52**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	91.21 / 91.05 / 90.98	**91.08 ± 0.12**	1.91	+**0.15** (p=0.55)
CVC-ClinicDB Polyp (Binary polyp, HD95 in px)						
DA-TransUNet [6]	108.00	39.80	89.72 / 89.40 / 89.29	89.47 ± 0.23	14.49	–
ApproxDA-TransUNet (Ours)	**10.78**	**9.00**	90.32 / 89.37 / 89.94	**89.88 ± 0.48**	**12.74**	+**0.41** (p=0.20)
ISIC 2018 Dermoscopy (Binary skin lesion, HD95 in px)						
DA-TransUNet [6]	108.00	39.80	88.08 / 88.48 / 88.35	88.30 ± 0.20	**14.15**	–
ApproxDA-TransUNet (Ours)	**10.78**	**9.00**	88.43 / 88.82 / 88.27	**88.51 ± 0.28**	14.83	+**0.20** (p=0.29)
FeTA 2.4 Fetal Brain MRI (7 tissue classes, HD95 in mm)						
DA-TransUNet [6]	107.95	30.21	80.93 / 79.61 / 80.23	80.26 ± 0.66	**2.17**	–
ApproxDA-TransUNet (Ours)	**10.75**	**6.92**	81.66 / 81.08 / 80.41	**81.05 ± 0.63**	2.85	+**0.80** (p=0.17)
Combined Multi-Benchmark Aggregate (15 Seed Pairs Across 5 Modalities)						
Positive Seed Pairs: **10 / 15**						Mean Paired Gain: +**0.37** pp (Wilcoxon $p = 0.010$)
Parameter Reduction: **90.0%** (107.95M → 10.75M)						Compute Reduction: **77.1%** (30.21G → 6.92G)

only on CVC-ClinicDB (12.74 vs. 14.49 px) and higher on the other four datasets (significantly only on FeTA, $p = 0.04$, see the case audit below).

4.3. FeTA 2.4 Benchmark: 2D and 3D Comparisons

Table 3 places ApproxDA-TransUNet among six 2D models trained under the same protocol and three 3D references:

- **CNN and Swin baselines**: ApproxDA-TransUNet (81.05%) is above U-Net (+1.85 points, $p = 0.018$), U-Net++ (+2.04, $p = 0.021$), Attention U-Net (+1.94, $p = 0.073$), and Swin-UNet (+2.46, $p = 0.059$).
- **Transformer hybrids**: it is 0.07 points below TransUNet, a small but consistent gap (lower for all three seeds, $p = 0.036$), with a tenth of the parameters and under a quarter of the FLOPs, and 0.80 points above DA-TransUNet ($p = 0.17$).
- **3D references**: its mean Dice lies between 3D UX-Net (80.01%) and MedNeXt-M (81.22%). The 3D numbers come from a single seed with validation-based checkpoint selection, so this comparison is indicative only.

Table 3: Tissue-level Dice (%) on the FeTA 2.4 test set (8 subjects). 2D models: three seeds under the protocol of Section 4, last epoch; per-tissue values are three-seed means, and p is a paired t-test against ApproxDA-TransUNet. FLOPs are per 224×224 slice. 3D models: our reproduction under the EffiDec3D protocol, one seed, checkpoint selected on the validation set; per-tissue values were not recorded, and their cost is given per 96^3 patch in multiply–accumulates (GMac), so it is not directly comparable with the 2D FLOPs.

Model	Type	Params (M)	Cost	eCSF	GM	WM	LV	CBM	SGM	BS	Mean DSC (%)	p vs. ours
U-Net [1]	2D	31.04	36.9 GFLOPs	73.8	67.7	84.4	84.1	**82.5**	83.1	78.9	79.21 ± 0.34	0.018
Attention U-Net [4]	2D	31.39	37.7 GFLOPs	73.5	66.8	84.5	84.4	82.2	83.5	78.9	79.11 ± 0.34	0.073
U-Net++ [2]	2D	9.16	26.5 GFLOPs	73.3	67.4	84.1	83.8	82.3	83.3	78.9	79.01 ± 0.16	0.021
Swin-UNet [11]	2D	27.17	6.2 GFLOPs	74.1	68.6	83.9	83.6	81.1	84.7	74.0	78.60 ± 0.46	0.059
TransUNet [3]	2D	105.28	29.3 GFLOPs	74.5	68.0	87.7	**88.9**	82.2	**85.0**	**81.4**	**81.12 ± 0.63**	0.036
DA-TransUNet [6]	2D	107.95	30.2 GFLOPs	73.9	68.7	85.2	87.4	81.9	84.1	80.6	80.26 ± 0.66	0.17
ApproxDA (Ours)	2D	**10.75**	6.9 GFLOPs	**75.5**	**71.3**	**87.9**	87.9	81.8	83.7	79.3	81.05 ± 0.63	–
3D reference models (our reproduction, EffiDec3D split, one seed, validation-selected checkpoint)												
SwinUNETR [13]	3D	62.19	308.0 GMac	–	–	–	–	–	–	–	78.38	–
3D UX-Net [26]	3D	53.01	578.5 GMac	–	–	–	–	–	–	–	80.01	–
MedNeXt-M [27]	3D	17.56	106.1 GMac	–	–	–	–	–	–	–	81.22	–

- **Tissues**: relative to DA-TransUNet, the gains are in grey matter (+2.6), white matter (+2.7), and external CSF (+1.6); cerebellum and deep grey matter are on par (−0.1, −0.4), and the brainstem is lower (−1.3).

Case Audit of FeTA HD95 (Sub-052 Outlier). ApproxDA-TransUNet has a higher mean HD95 on FeTA than DA-TransUNet (2.85 vs. 2.17 mm, Table 2). The difference comes from a single subject, **sub-052**, which has severe ventriculomegaly (lateral-ventricle to white-matter volume ratio 2.29, against a training median of 0.16 and a training maximum of 1.81) and is therefore outside the training distribution. On this subject, ApproxDA-TransUNet predicts a few small cerebellum regions (the largest has 34, 170, and 266 voxels for the three seeds) about 50 mm away from the true cerebellum, which raises the cerebellum HD95 to 28–50 mm and the subject's mean HD95 to 8.88, 8.89, and 13.65 mm (DA-TransUNet: 2.85, 4.43, 4.02 mm). Excluding sub-052, the mean HD95 over the other seven subjects is 1.76 **mm** for ApproxDA-TransUNet and 1.94 mm for DA-TransUNet.

4.4. Qualitative Visual Evaluation

[TODO: rewrite this paragraph after the figure is regenerated.]

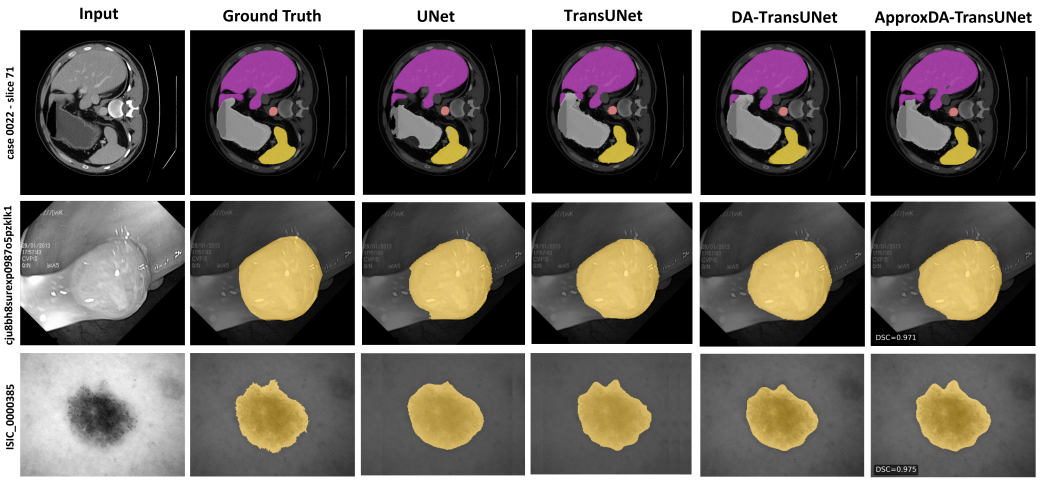


Figure 3: Qualitative segmentation comparisons. [TODO: this is the workshop figure (ViT-B ApproxDA model; the middle row is Kvasir-SEG, not CVC-ClinicDB; no ACDC or FeTA rows). Regenerate it with the final model on the five datasets, showing ground truth, DA-TransUNet, and ApproxDA-TransUNet.]

5. Ablation Studies

Single-axis ablations are run on Synapse with three seeds (1234, 0, 1). The reference is ApproxDA-TransUNet ($80.57 \pm 0.19\%$ Dice, HD95 24.52 voxels); each variant changes exactly one factor.

5.1. Block and Attention Operator

- **The block is necessary**: removing all ApproxDA blocks lowers Dice by 2.76 **points** for every seed ($p = 0.008$) and raises HD95 from 24.52 to 29.12; the largest drops are in the gallbladder and pancreas.

- **The attention operator is not**: exact global dual attention ($M = 112$, $r = 0$, $G = 1$; -0.01, $p = 0.97$), exact attention inside windows ($r = 0$; $+0.19$), and the convolutional path alone with both attention operators disabled ($+0.21$, $p = 0.57$) all match the baseline. The exact global variant costs 14% more FLOPs and about twice the training time per run (94 vs. 45 minutes). The accuracy of the block therefore comes from its convolutional path, consistent with the gradient analysis in Section 6.1, and the approximation does not cost accuracy.

Table 4: Single-axis ablations on Synapse (seeds 1234 / 0 / 1, 224×224, 9 classes). The baseline is ApproxDA-TransUNet ($M = 28$, $r = 32$, $G = 8$, $g = 0.5$, LightViT-T, slim decoder). HD95 in voxels. Paired differences and p-values (paired t-test) are relative to the baseline. FLOPs are counted on CPU with explicit attention products.

Ablation Variant	Params (M)	FLOPs (G)	Mean DSC (%)	HD95	Paired Diff. (p)
ApproxDA-TransUNet baseline	**10.75**	**6.92**	**80.57 ± 0.19**	**24.52**	–
Block presence and attention operator					
w/o ApproxDA blocks	10.20	6.3	77.82 ± 0.33	29.12	−2.76 (0.008)
Convolutional path only (attention disabled)	10.75	6.92	80.78 ± 0.72	21.80	+0.21 (0.57)
Exact global dual attention (M=112, r=0, G=1)	10.65	7.90	80.56 ± 0.62	23.64	−0.01 (0.97)
Window size M					
$M = 112$ (global window at every stage)	12.25	6.92	80.07 ± 0.98	28.99	−0.50 (0.54)
Projection rank r					
$r = 0$ (exact attention inside each window)	10.65	7.03	80.76 ± 0.15	21.12	+0.19 (0.44)
$r = 64$	10.85	6.93	79.64 ± 0.40	26.85	−0.93 (0.11)
Channel groups G					
$G = 1$ (un-grouped CAM)	10.75	6.92	81.12 ± 0.77	23.48	+0.55 (0.25)
$G = 4$	10.75	6.92	79.91 ± 0.57	26.69	−0.67 (0.10)
Encoder					
R50 third stage instead of LightViT-T	15.42	8.7	78.81 ± 0.27	27.71	−1.76 (0.007)
ViT-B/16 instead of LightViT-T	100.47	26.0	78.52 ± 1.64	26.01	−2.05 (0.14)
One strided 3×3 convolution instead of LightViT-T	3.53	6.3	80.55 ± 0.36	23.74	−0.02 (0.94)
One strided convolution, w/o ApproxDA blocks	2.98	5.7	78.98 ± 0.17	34.39	−1.60 (0.001)
LightViT-S instead of LightViT-T	20.50	8.0	80.31 ± 0.63	23.62	−0.26 (0.43)
Bottleneck placement and decoder					
Bottleneck block before LightViT-T	11.06	7.2	79.47 ± 0.36	26.10	−1.10 (0.008)
Full decoder ($256 \to 128 \to 64 \to 16$)	14.24	10.4	79.60 ± 1.64	25.63	−0.97 (0.40)

5.2. Approximation Hyperparameters

- **Window size and rank**: a global window ($M = 112$, -0.50, $p = 0.54$) and a higher rank ($r = 64$, -0.93, $p = 0.11$) are not significantly different from the defaults; with three seeds on Synapse, differences below about one point cannot be resolved.
- **Channel groups**: un-grouped CAM ($G = 1$) is 0.55 points higher ($p = 0.25$). With $G = 8$, CAM degenerates to a scaling in the bottleneck and the 1/2 skip (Section 3); $G = 1$ avoids this at no extra cost.
- **Channel groups** $G = 4$: -0.67 ($p = 0.10$), not significant.
- **Encoder**: replacing LightViT-T with the third ResNet-50 stage (the TransUNet stem, 15.42M parameters) lowers Dice by 1.76 points for every seed ($p = 0.007$) while adding 4.7M parameters. ViT-B/16 is also lower (-2.05, $p = 0.14$), and LightViT-S is not better (-0.26, $p = 0.43$).
- **Block placement**: applying the bottleneck block before LightViT-T, on the $28 \times 28 \times 512$ stage-2 output as in DA-TransUNet's ordering,

lowers Dice by 1.10 points for every seed ($p = 0.008$) and adds 0.31M parameters. The full decoder of DA-TransUNet is not better than the slim decoder (-0.97, $p = 0.40$) at 1.3 times the parameters and 1.5 times the FLOPs.

- **Transformer versus blocks**: crossing the bottleneck module (one strided convolution or LightViT-T) with the presence of ApproxDA blocks gives a 2×2 design. The blocks help in both cases ($+1.57$ points with the convolution, $p = 0.034$; $+2.76$ with LightViT-T, $p = 0.008$) and lower HD95 by 10.7 and 4.6 voxels. LightViT-T, in contrast, does not help: without blocks it is 1.16 points below the convolution ($p = 0.049$), and with blocks the two are equal (80.57 vs. 80.55, $p = 0.94$). [TODO (author decision): on Synapse the 3.53M convolutional variant with ApproxDA blocks matches the final model; decide how the paper motivates LightViT-T.]

6. Discussion and Mechanistic Insights

This section analyzes the optimization of dual attention under SGD, discusses left–right errors caused by mirror augmentation, reports deployment costs, and outlines limitations.

6.1. Optimization Diagnostics: Zero-Initialized Residual Scales

A common convention in dual attention [5, 6] is the zero-initialized residual scaling:

$$\mathbf{Y}_{\mathrm{PAM}} = \alpha \mathbf{E}_{\mathrm{PAM}} + \mathbf{X}, \quad \alpha_{\mathrm{init}} = 0, \tag{12}$$

and similarly $\beta_{\mathrm{init}} = 0$ for CAM. Under SGD this initialization leaves the position-attention weights almost untrained.

By the chain rule, the gradient with respect to projection weights $\mathbf{W}_Q$ is:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_Q} = \frac{\partial \mathcal{L}}{\partial \mathbf{Y}_{\mathrm{PAM}}} \cdot \left(\alpha \frac{\partial \mathbf{E}_{\mathrm{PAM}}}{\partial \mathbf{W}_Q} \right). \tag{13}$$

At initialization, because $\alpha = 0$, $\partial \mathcal{L} / \partial \mathbf{W}_Q$ evaluates strictly to zero. The only way out is through α itself, whose gradient $|\partial \mathcal{L} / \partial \alpha| = |\langle \partial \mathcal{L} / \partial \mathbf{Y}_{\mathrm{PAM}}, \mathbf{E}_{\mathrm{PAM}} \rangle|$ we measured at 4×10^{-6} to 4×10^{-5}, about four orders of magnitude below the gradients of the decoder convolutions (≈ 0.29).

Empirical verification. With weight decay alone and no gradient, a weight shrinks by $(1 - \eta\lambda)^T \approx 0.84$–$0.86$ over training. In our SGD-trained models with the ViT-B/16 encoder (the configuration of our earlier study [7]), the ratio of the final $\max |w|$ of the PAM query, key, value, and projection weights to their initialization bound was 0.84–0.86, i.e., they were never updated by gradient descent, and setting $\alpha = 0$ at test time changed at most 0.0002% of the predicted pixels. The effect is specific to SGD and PAM: with Adam (CVC-ClinicDB) the attention weights did train, and the CAM scale β moved away from zero on Synapse and CVC-ClinicDB.

The final model. The same holds for ApproxDA-TransUNet. In the twelve SGD-trained models (Synapse, ACDC, ISIC 2018, and FeTA, three seeds each; 48 blocks), the learned scales stay at $|\alpha| \leq 0.011$ and $|\beta| \leq 0.06$. Switching both attention operators off at test time, so that each branch passes its compressed input through unchanged, changes 0.006–0.016% of the predicted voxels on the Synapse test set and the Dice by +0.01, +0.00, and +0.09 points for the three seeds. With Adam (CVC-ClinicDB), weight decay drives the PAM query and key weights to zero in three of the four blocks, and only the 1/8 skip block learns a non-zero α ($|\alpha| \approx 0.27$).

What the block contributes. Together with the ablations in Section 5, where removing the block costs 2.76 points but replacing its attention operator by an exact, approximate, or disabled one changes nothing, this indicates that the contribution of dual-attention blocks in this setting comes from their convolutional path ($C \to C/16 \to C$) rather than from the spatial attention matrices. It also explains why the approximation ($M = 28$, $r = 32$, $G = 8$) costs no accuracy.

6.2. Anatomical Symmetries and Data Augmentation

In an error decomposition on the Synapse test set (with the earlier ViT-B/16 model), about 87% of the misclassified voxels were boundary or extent errors between an organ and the background; confusions between organs were mostly whole-organ misidentifications, about 9% of all errors, chiefly stomach versus spleen and left versus right kidney, concentrated in a few test cases.

The left–right confusions are consistent with the standard TransUNet augmentation: `random_rot_flip` follows a random 90° rotation with an axis flip without swapping the left and right labels, so about half of the augmented slices show mirrored anatomy. Laterality-aware augmentation should be considered for bilateral structures. [TODO: the quantitative effect of removing

Table 5: Deployment profile on an NVIDIA RTX 5090 (fp32, 224×224 input, 9 classes, idle GPU). Memory is the peak allocated by PyTorch, including weights and cuDNN workspace; training uses batch size 24 with SGD.

Metric	**DA-TransUNet**	**ApproxDA-TransUNet (Ours)**
Parameters	107.95 M	**10.75 M** (−90.0%)
GFLOPs per slice	30.21	**6.92** (−77.1%)
Checkpoint size (FP32)	432 MB	**43.5 MB**
Peak training memory (batch 24)	11.52 GiB	**4.85 GiB** (−58%)
Peak inference memory (batch 1)	2.59 GiB	**1.09 GiB** (−58%)
Throughput, batch 32 (slices/s)	555	**1415** (2.55×)
Latency, batch 1	**6.5 ms**	9.6 ms

mirroring was measured on AMOS, which is not reported; on Synapse the no-mirror model still had about 9% whole-organ errors. Either add a Synapse measurement or keep this paragraph qualitative.]

6.3. Computational Efficiency and Clinical Deployability

Table 5 summarizes the deployment cost. ApproxDA-TransUNet needs 58% less GPU memory for training and inference and stores a ten-times smaller checkpoint. Because volumes are segmented slice by slice in batches, throughput is the relevant speed measure, and it is 2.55 times higher. At batch size 1, however, the lightweight model is slower (9.6 vs. 6.5 ms): its many small LightViT and attention kernels leave the GPU underutilized, so the FLOP reduction does not translate into lower single-slice latency.

6.4. Limitations and Future Work

ApproxDA-TransUNet processes 3D scans on a 2D planar basis. It therefore cannot use inter-slice context, which matters, for example, at the cranial and caudal ends of organs. Extending ApproxDA blocks to lightweight 2.5D or anisotropic 3D formulations presents a promising avenue for volumetric research.

7. Conclusion

In this paper, we introduced **ApproxDA-TransUNet**, a lightweight dual-attention segmentation network that combines a truncated ResNet-50 stem, LightViT-T, approximated dual-attention blocks ($M = 28$, $r = 32$, $G = 8$) on the bottleneck and all skips, and a slim decoder. It uses

90.0% fewer parameters (10.75M vs. 107.95M) and 77.1% fewer FLOPs (6.92 vs. 30.21 GFLOPs) than DA-TransUNet. Over three seeds on five benchmarks (Synapse, ACDC, CVC-ClinicDB, ISIC 2018, and FeTA 2.4), its Dice matches or slightly exceeds that of DA-TransUNet (mean paired gain +0.37 points, Wilcoxon $p = 0.010$), while HD95 is higher on four of the five datasets; on FeTA 2.4 it is within 0.1 points of TransUNet.

Ablations show that the ApproxDA block is necessary but that its attention operator can be approximated, made exact, or disabled without changing accuracy, and our gradient analysis of zero-initialized residual scales under SGD explains this. In deployment it needs 58% less GPU memory and processes 2.55 times more slices per second at batch size 32, although single-slice latency at batch size 1 is not lower.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Authorship Contribution Statement

Author One: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - Original Draft, Visualization. **Author Two**: Software, Validation, Data curation, Writing - Review & Editing. **Author Three**: Supervision, Project administration, Resources, Funding acquisition, Writing - Review & Editing.

Data and Code Availability

All datasets used in this study (Synapse BTCV, ACDC, CVC-ClinicDB, ISIC 2018, and FeTA 2.4) are publicly accessible benchmark repositories. The complete source code and pretrained model checkpoints for ApproxDA-TransUNet will be made publicly available upon publication.

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